THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The molecular/formula unit mass of ammonia, NH_3 is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 19 u **(b)** 34 u **(c)** 15 u **(d)** 17 u

2. The molecular/formula unit mass of calcium carbonate, $CaCO_3$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 200 u **(b)** 100 u **(c)** 98 u **(d)** 102 u

3. If 2.4 g of carbon reacts completely with oxygen (C : O = 12 : 32), the mass of CO_2 formed is: 1

(a) 8.8 g **(b)** 6.2 g **(c)** 4.4 g **(d)** 10.4 g

4. Carbon monoxide has C : O = 3 : 4. The mass of oxygen that combines with 9 g carbon is: 1

(a) 6 g **(b)** 12 g **(c)** 16 g **(d)** 9 g

5. The chemical formula of magnesium oxide is: 1

(a) Mg_2O_2 **(b)** MgO_2 **(c)** MgO **(d)** Mg_2O

6. An atom of calcium has atomic number 20 and mass number 40. The number of neutrons in it is: 1

(a) 21 **(b)** 40 **(c)** 19 **(d)** 20

7. The electronic configuration (K, L, M, N shells) of the sodium atom (Z = 11) is: 1

(a) 2, 9 **(b)** 2, 8, 1 **(c)** 8, 2, 1 **(d)** 2, 8, 2

8. The electronic configuration (K, L, M, N shells) of the calcium atom (Z = 20) is: 1

(a) 2, 8, 8, 2 **(b)** 2, 8, 9, 1 **(c)** 2, 9, 9 **(d)** 2, 8, 8, 3

9. The valency of chlorine (Z = 17) is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 4

10. The maximum number of electrons that the shell with n = 4 can hold is: 1

(a) 16 **(b)** 64 **(c)** 32 **(d)** 34

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define a molecule. Give one example each of a molecule of an element and of a compound. 2

12. How is a covalent bond formed? Give one example. 2

13. Define atomic number and mass number. 2

14. An atom has atomic number 26 and mass number 56. Find the number of protons, electrons and neutrons. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain the formation of a water molecule by sharing of electrons. 3

16. An atom has 18 neutrons and mass number 35. Find its atomic number, name and configuration. 3

17. Explain the formation of sodium chloride from its atoms. 3

OR

Explain the formation of magnesium chloride. 3

18. An atom has atomic number 26 and 56 nucleons. Find the number of neutrons, electrons and its symbol. 3

OR

Explain with the examples of sodium, oxygen and carbon how the valency of an element is determined. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A student compares sodium chloride and sugar. 5

(i) Which conducts electricity in aqueous solution?
(ii) Why?
(iii) Compare their melting points.
(iv) Classify them as ionic or covalent.

OR

(i) Write the chemical formulae of calcium chloride, aluminium oxide, sodium sulfate, magnesium oxide and ammonia. 5
(ii) Name the rule used.
(iii) Name CO, CO_2, CS_2, N_2O_4, PCl_3.

20. (i) Define isotopes. 5
(ii) Name the three isotopes of hydrogen and give their compositions.
(iii) Why are the chemical properties of isotopes similar but the physical properties different?

OR

Case study: Natural chlorine consists of isotopes of mass 35 u and 37 u in the ratio 3 : 1. 5
(i) Name the isotopes.
(ii) Find the simple average mass.
(iii) Find the weighted average atomic mass.
(iv) Why is the weighted value used?

END OF QUESTION PAPER Total: 40 Marks